



THE UNIVERSITY OF AUCKLAND
NEW ZEALAND

PRIVACY ENHANCING TECHNOLOGIES

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COMPSCI 316



Technologies that allow to utilise personal data while minimising privacy risks

Technologies that allow to utilise personal data while minimising privacy risks



PETs

- The concept of PETs was first discussed in a report published by the Information and Privacy Commissioner of Ontario, Canada (1995) "[Privacy-Enhancing Technologies: The Path to Anonymity](#)"
- From the end-user perspective : any technology that allow them to interact with digital systems in a privacy-preserving manner
- From the organisational perspective : any technology that allow them to develop a privacy-enhancing product or a service
- Every PET **does not** guarantee anonymisation, but they enhance data privacy by minimizing the personal data usage

Anonymity



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joyoftech.com

Anonymity

- The state of being unknown / undetectable
- Anonymisation – the technique of removing the link between data and the data subject

Example

Name	Age	Disability
Kate	21	Vision impairment



Name	Age	Disability
ILBer#	20-30	Vision impairment

- Is anonymous data governed by GDPR?
- Can you think of negative implications of anonymity in the digital world?

PETs

How do PETs achieve data protection

- Transform data or derive new data
- Hide or shield data
- Split datasets

PETs

Some Popular PETs

- Pseudonymisation
- K-Anonymity
- Differential Privacy
- Federated Learning
- Homomorphic Encryption
- Zero-Knowledge Proof
- Synthetic Data

There are more.....any technique that fits the definition

PETs

Some Popular PETs

- Pseudonymisation
 - K-Anonymity
 - Differential Privacy
-
- Federated Learning
 - Homomorphic Encryption
 - Zero-Knowledge Proof
 - Synthetic Data

- Converting personal data in a way that data can no longer be linked to a person, **without using additional information**
- Why it is different from anonymisation?
- 2 Steps (Most of the time)
 1. Replace direct identifier(s) with **an** artificial identifier -> **Always**
 2. Save actual identifiers in a separate secure table (mapping table) -> **Depends on technique used in 1**
- Mapping table and pseudonimised tables are linked through the pseudonym column

PETs

Pseudonymisation

Name	Age	ZIP	Gender	Disease
Jhon	29	1981	M	Cancer
Kate	22	1980	F	Infection
Alice	35	1975	F	AIDS
Ellen	38	1978	F	AIDS
Mary	41	1974	F	Cancer

Raw Data



ID	Age	ZIP	Gender	Disease
P001	29	1981	M	Cancer
P002	22	1980	F	Infection
P003	35	1975	F	AIDS
P004	38	1978	F	AIDS
P005	41	1974	F	Cancer

Pseudonymised Data

ID	Name
P001	Jhon
P002	Kate
P003	Alice
P004	Ellen
P005	Mary

Mapping Table

← Needs Security

PETs

Generate the pseudonyms

- Counter

Replace identifiers with a number or a string

Increment by 1 for each identifier

Each identifier has a unique pseudonym : **No collisions**

ID	Name
P001	Jhon
P002	Kate
P003	Alice

ID	Name
0	Jhon
1	Kate
2	Alice

ID	Name
10@gmail.com	jhon@gmail.com
11@abc.ac.nz	bob@abc.ac.nz
12@de.nz	mary@de.nz

PETs

Generate the pseudonyms

- Random Number Generator (RNG)

Produce values within a set

Collision issue

```
import random

for i in range(10):
    print(random.randint(0,9))
```

ID	Name
9	Jhon
6	Kate
3	Alice
9	Mary
4	Dirk

PETs

Generate the pseudonyms

- Cryptographic Hashing

Create a fixed-length output from an arbitrary-length input

Cannot be reversed to the original value

No collisions for different inputs

```
import hashlib

# identifiers
names = ["Jhon", "Kate", "Alice", "Mary", "Dirk"]

# hashing using SHA256
for name in names:
    result = hashlib.sha256(str.encode(name))
    print(result.hexdigest())
```

ID	Name
ee0ace6e8f5dc17dc271cb6e7c0cdc2de39f84c84541461b95ed7c59414becf0	Jhon
1a5d06a170dde413475957ca2b63396aa5e8fbecb7d379fcb668da3753fca5b6	Kate
3bc51062973c458d5a6f2d8d64a023246354ad7e064b1e4e009ec8a0699a3043	Alice
aebac53c46bbeff10fdd26ca0e2196a9bfc1d19bf88eb1efd65a36151c581051	Mary
764e8ab23aba697ce8365352a91b8e3c57b8f6672c6873f7d9de17f254a31cce	Dirk

PETs

Generate the pseudonyms

- Message Authentication Code (MAC)

Create a fixed-length output from an arbitrary-length input

A secret key is needed

No collisions for different inputs

```
import hmac
import hashlib

# identifiers
names = ["Jhon", "Kate", "Alice", "Mary", "Dirk"]

# secret key
key = "c189026b-62b0-43946-8a38*e911"

# using HMAC (Keyed Hash MAC)
for name in names:
    result = hmac.new(str.encode(key), str.encode(name), hashlib.sha256)
    print(result.hexdigest())
```

ID	Name
9c2975017f9a1c469c35e44346b1b728b5a260c86bbfb5e7cac1c941f63af3d4	Jhon
5d5993a32fb5b290791bcd87fdda58ef0c71b60ea104a43eb7c38a8260b0fa2	Kate
b09a9911748cfe38306191122d8f3c2b363cac875b8c1920e7406b33628880f7	Alice
99cd4f78b2fdbdd01bd5d0c44651ab4201a64cf97ae576ac4b813c3bb18cb2d2	Mary
f7ed755df953d154a9953d521adfc6956f7bd2ef31be6e4374158a7f058f513	Dirk

PETs

Generate the pseudonyms

- Encryption

Two way

A secret key is needed

No collisions

Deterministic encryption on different values

Non-deterministic encryption

```
from Crypto.Cipher import AES
from Crypto.Random import get_random_bytes

# identifiers
names = ["Jhon", "Kate", "Alice", "Mary", "Dirk"]

#secret key
key = get_random_bytes(16)

cipher = AES.new(key, AES.MODE_CTR)

#using AES128 encryption
for name in names:
    ciphertext = cipher.encrypt(str.encode(name))
    print(ciphertext.hex())
```

ID	Name
6ae9b676	Jhon
4672d45f	Kate
87967253b7	Alice
dbaa6bf9	Mary
018f1600	Dirk

- Does all Pseudonymisation techniques require a mapping table?
- Scalability
- What if the same identifier is repeating in multiple tables?
 - Ex: A university student can also be in the university employee table
 1. Deterministic Pseudonymisation
 2. Document Randomised Pseudonymisation
 3. Fully Randomised Pseudonymisation
- Can we represent multiple identifiers from one pseudonym
 - {name, driver_licence , email}

_____ can escape GDPR. Which is/are correct?

0%



Anonymisation

0%



Pseudonymisation



1



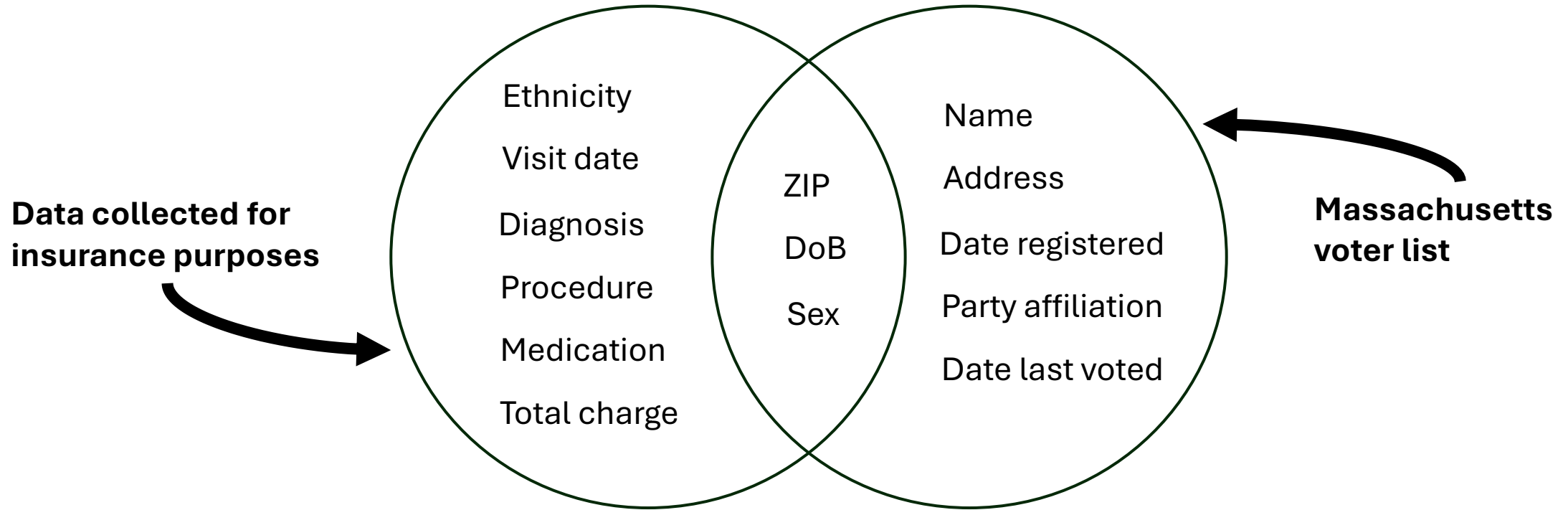
0



0/50



Linkage Attacks



“87% of the US population can be uniquely identified by only {ZIP, DoB, Sex} data fields”

- Latanya Sweeney

- Making an individual indistinguishable from $k-1$ individuals in the same dataset : “[hiding in a crowd](#)”
- It is achieved through generalisation or suppression of quasi-identifiers
- Terminology
 - Quasi-identifiers - data attributes, excluding PII, that in combination can uniquely identify a person
(ex: [age, zip code, race, and gender of a patient](#))
 - Generalisation - replacing an individual data value with a more general value
(ex: [age 20 replaced by age range 20-30](#))
 - Suppression - removing data values in a dataset to achieve anonymity
(ex: [Zip 1234 replaced by 123*, removing an entire column](#))
- K-anonymity provided syntactic anonymisation

PETs

K-Anonymity

ID	Age	Zip	Gender	Disease
P001	22	2141	M	Cancer
P002	24	2141	F	Infection
P003	31	2138	F	AIDS
P004	32	2139	F	AIDS
P005	41	2243	M	Cancer
P006	41	2245	M	Infection
P007	48	6534	M	Infection

Quasi Identifiers Sensitive Data

2-Anonymous Data (k=2)

ID	Age	Zip	Gender	Disease
P001	21 – 30	2141	Human	Cancer
P002	21 – 30	2141	Human	Infection
P003	31 – 35	213*	F	AIDS
P004	31 – 35	213*	F	AIDS
P005	41 - 50	*	M	Cancer
P006	41 - 50	*	M	Infection
P007	41 - 50	*	M	Infection

Generalisation
Suppression

k = 2 => **at least** 2 rows in each group

Is it still possible to know what disease a patient had if you know background details of that patient?

Alice is a 32-year-old female who lives in area 2138.....

Kate lives in area of 2141, and she purchased antibiotics on her way to home after the hospital visit...

2-Anonymous Data (k=2)

ID	Age	Zip	Gender	Disease
P001	21 – 30	2141	Human	Cancer
P002	21 – 30	2141	Human	Infection
P003	31 – 35	213*	F	AIDS
P004	31 – 35	213*	F	AIDS
P005	41 - 50	*	M	Cancer
P006	41 - 50	*	M	Infection
P007	41 - 50	*	M	Infection

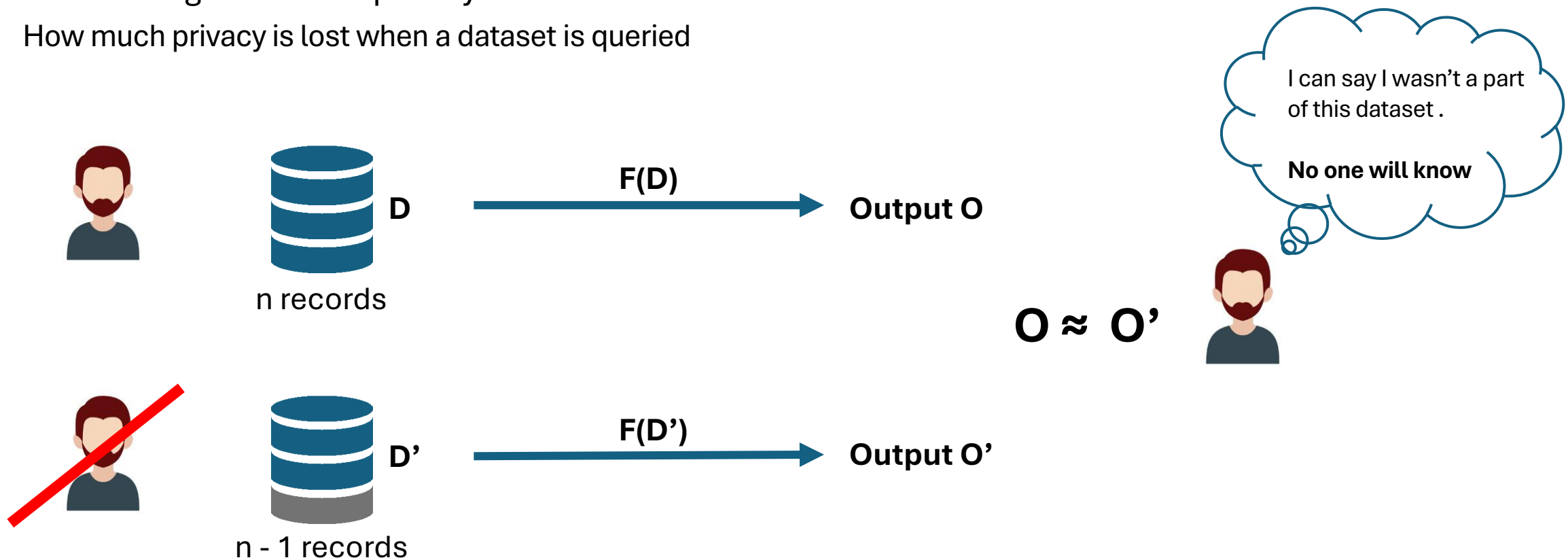
PETs

Differential Privacy (DP)

Adding random noise to the data so that it is hard to tell whether an individual is participating in a dataset or not

A mathematical guarantee of privacy

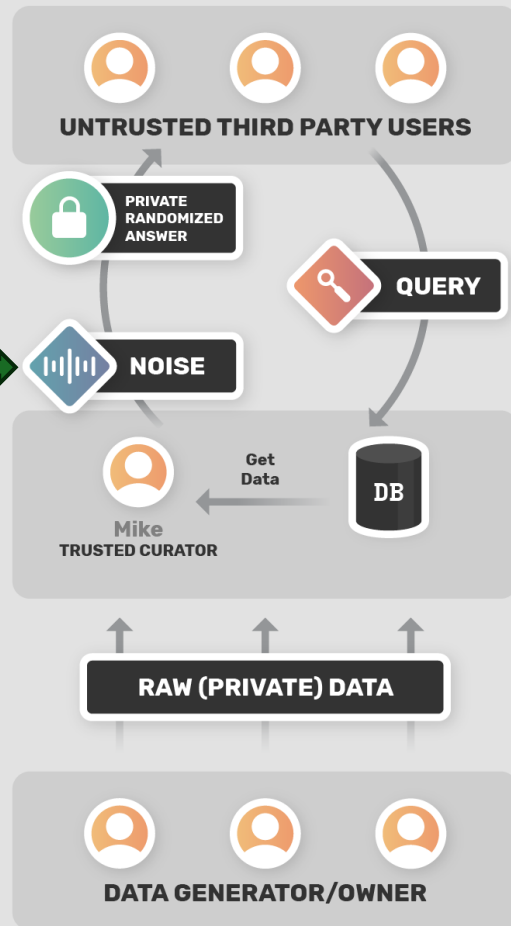
How much privacy is lost when a dataset is queried



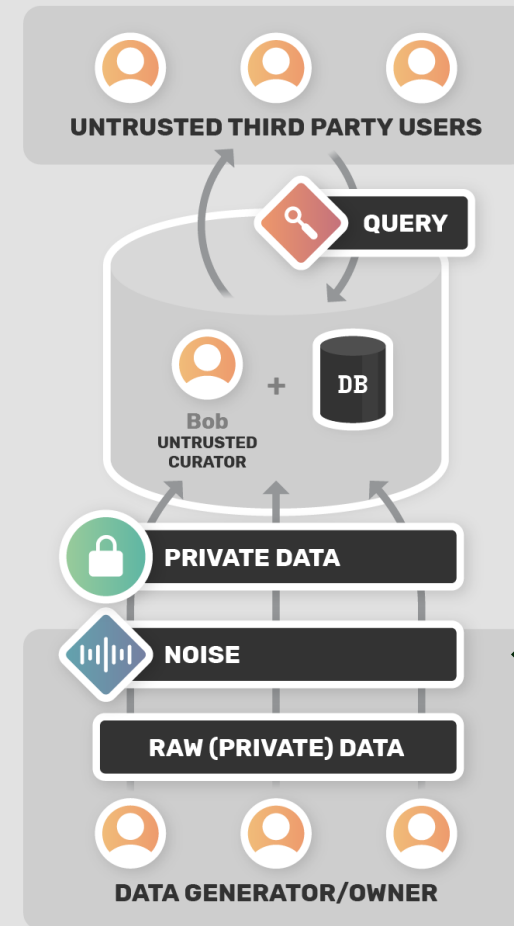
PETs

Differential Privacy (DP)

Global Differential Privacy



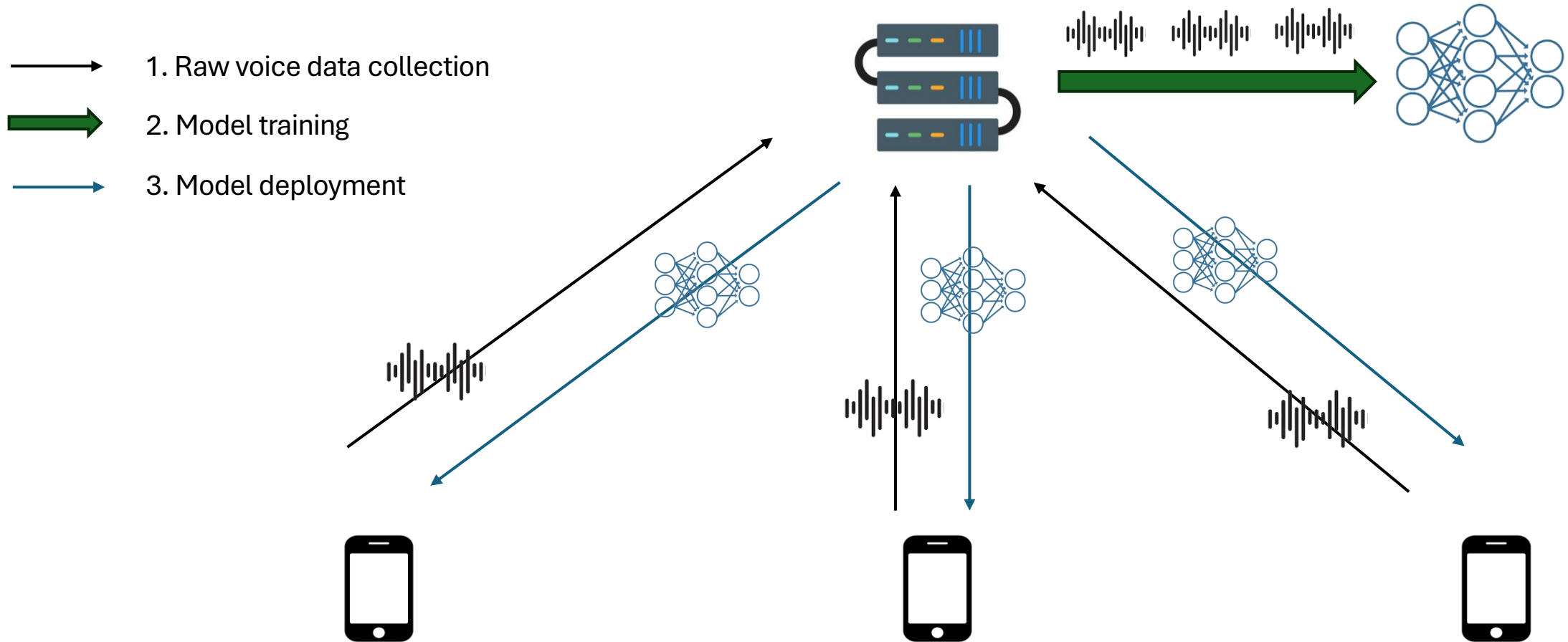
Local Differential Privacy



PETs

Federated Learning (FL)

Scenario : Apple using machine learning with **less privacy** to train Siri in identifying only the iPhone owner's voice



PETs

Federated Learning (FL)

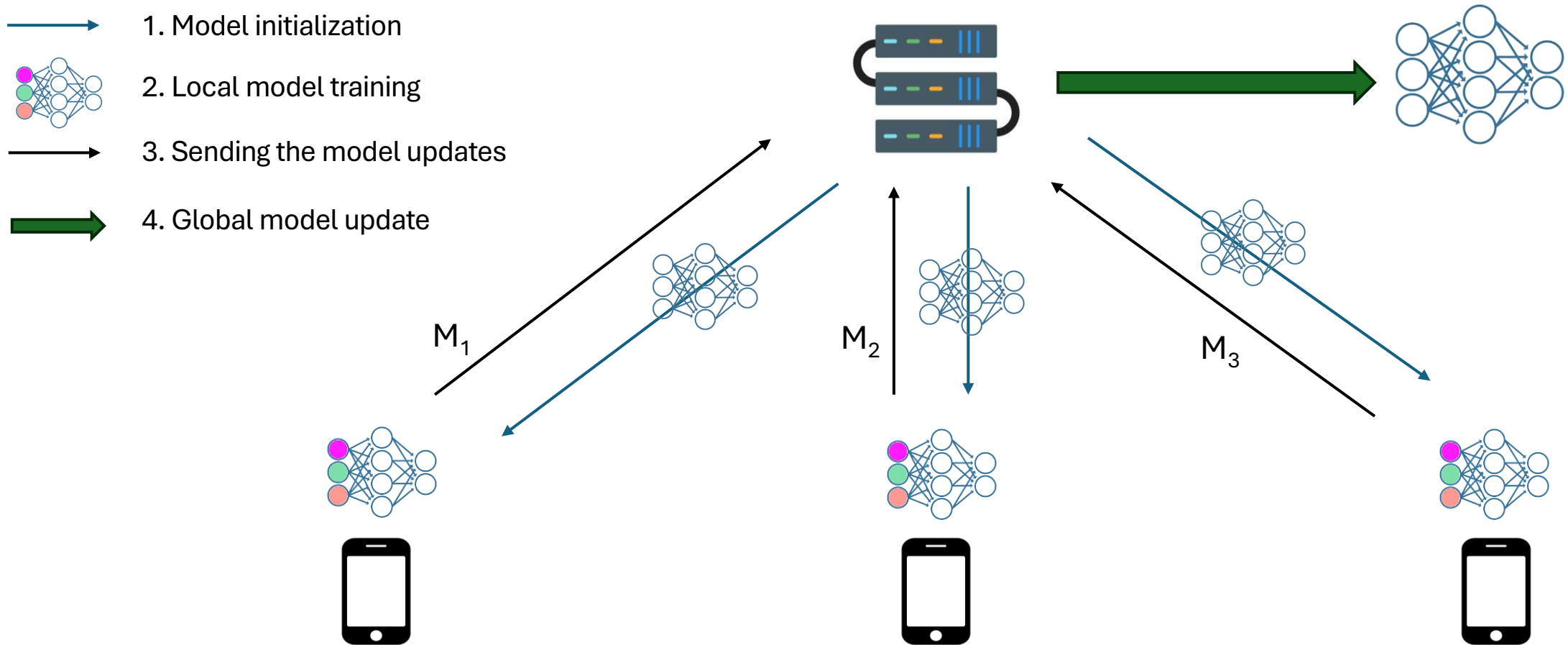
- Privacy preserving machine learning
- Data is not sent to a centralised location to train a machine learning model
- Setup
 - There is a central server
 - Nodes are connected to the central server (e.g., mobile phones)
 - Data is collected at the local nodes
 - Nodes can connect and disconnect from the server at any time

Data never leave the source !!

PETs

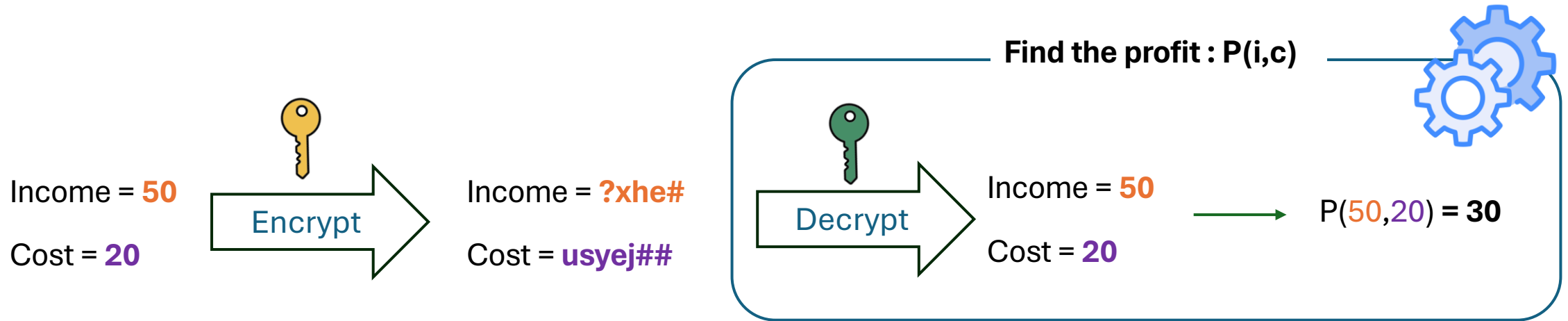
Federated Learning (FL)

Scenario : Apple using **privacy preserving** machine learning to train Siri in identifying only the iPhone owner's voice



PETs

Homomorphic Encryption (HE)



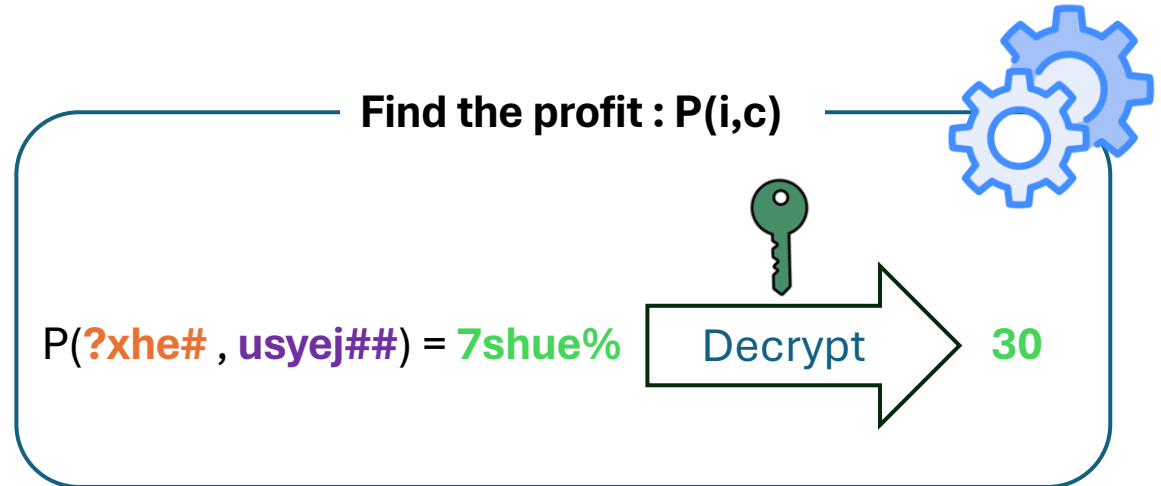
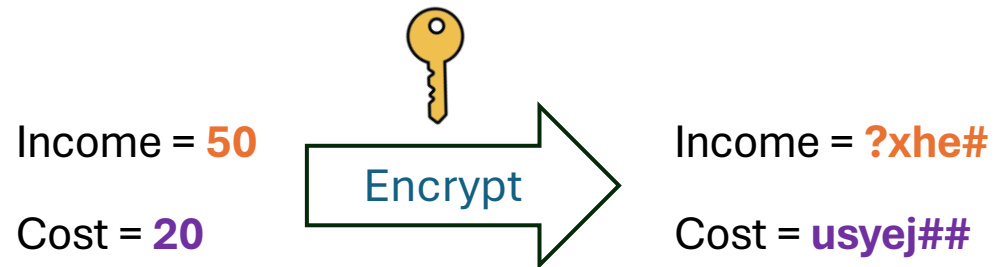
- If we want to process the encrypted data
 - The data must be reversed to the original format
 - Privacy violations are possible

PETs

Homomorphic Encryption (HE)

- What if we can perform operations on encrypted data without decrypting them

Homomorphic Encryption



- For arithmetic operations : addition and/or multiplication

PETs

Homomorphic Encryption (HE)

- Types of Homomorphic Encryption

Types	Operation (+,*)	Number of times
Partially	One	Unlimited
Somewhat	Both	Limited
Fully	Both	Unlimited

- Can we perform other arithmetic operations using + and *, if we have unlimited chances to use them?

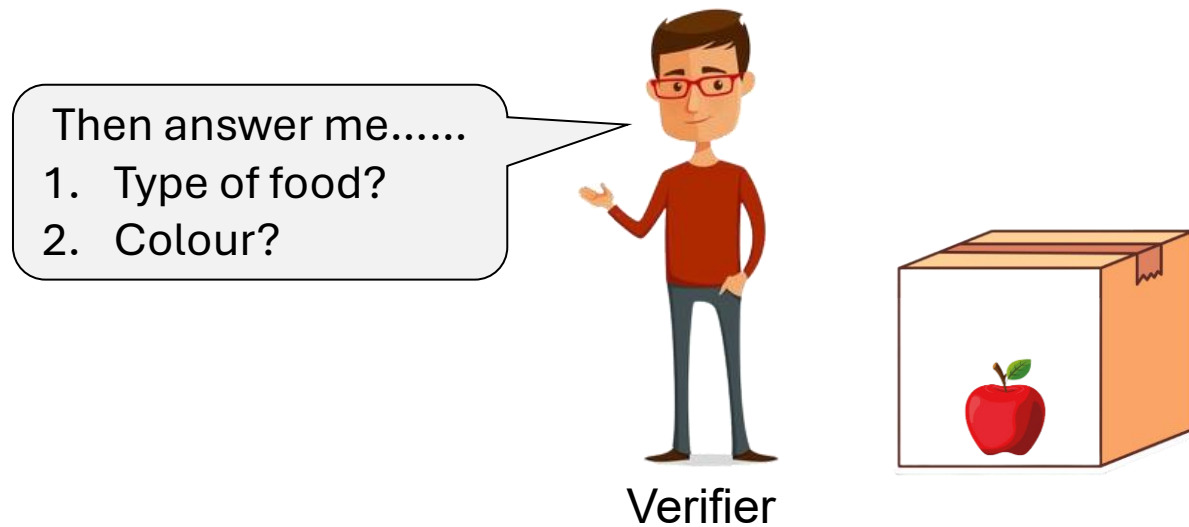
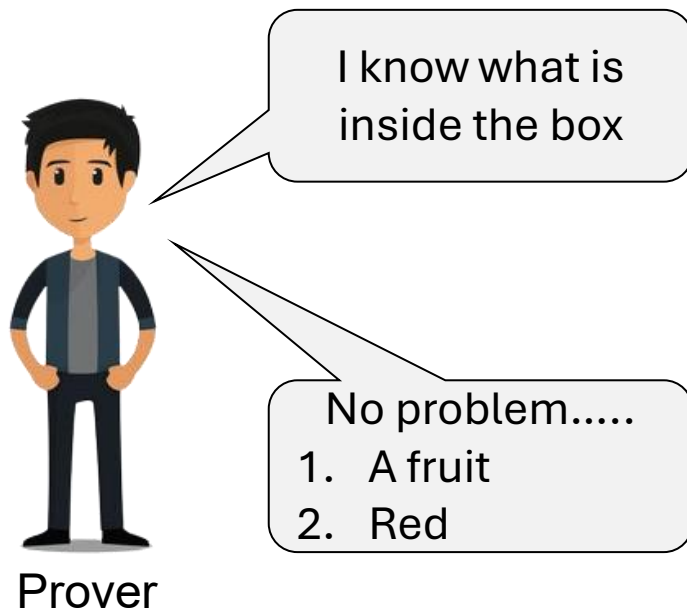
Almost

$$\begin{aligned} &4/2 \\ &= 4 + (-1 * 2) = 2 \\ &= 2 + (-1 * 2) = 0 \end{aligned} \quad \text{How many times} \Rightarrow 2 \text{ (answer)}$$

PETs

Zero Knowledge Proof (ZKP)

- Allow one party to prove the validity of a claim to another party without revealing the data bound to the claim
- A cryptographic technique



PETs

Zero Knowledge Proof (ZKP)

- There are 2 roles in ZKP
 1. **Prover** : wants to show they have knowledge of something
 2. **Verifier** : check the prover's claim

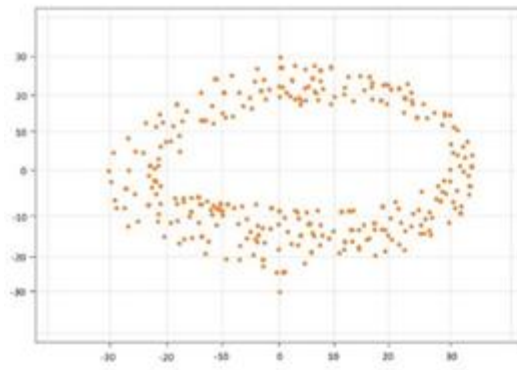
- **Proof size** : the amount of information passed between prover and verifier during the process

- There are 3 criteria to be met in ZKP
 1. **Completeness** : An honest prover always convince the verifier
 2. **Soundness** : Verifier can reject the false proofs shared by the dishonest prover
 3. **Zero-knowledge** : Prover knows nothing beyond the claim made by the prover

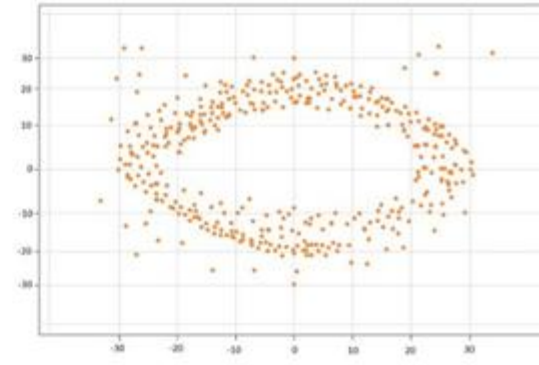
PETs

Synthetic Data

- Artificially generated data that mimics the statistical properties of real data



Original Data



Synthetic Data

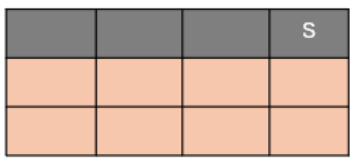
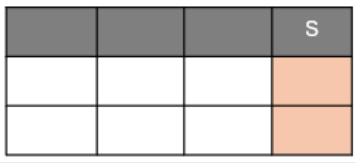
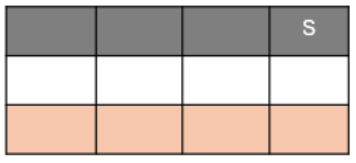
Navigating The Potential And Perils Of Synthetic Data In Healthcare (Shashank Agarwal 2024)

When do we need it?

PETs

Synthetic Data

- Types of synthetic data

Types	Synthetic Data Usage 
Fully	
Partially	
Hybrid	

			S

original dataset

S = sensitive data

- How to generate Synthetic Data?

- Statistical Distribution

- Using algorithms that can generate synthetic data using the statistical properties (ex: mean, variance etc) of the original data

- Generative Adversarial Networks

- A deep learning technique that use 2 neural networks to generate new data

- Generator Network – generating the new data

- Discriminator Network – Deciding how close the data to the real data

- Variational Autoencoders (VAEs)

- Encoded-decoded architecture

- VAE transforms data into a lower-level representation (encode) then try to regenerated it back (decode)

PETs

Canvas – More Than Privacy Protection

10:00

Start	Stop	Reset	mins: 10	secs: 0	type: None	▼
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Breaktime for PowerPoint by Flow Simulation Ltd.

Show Settings ☐

PETs

Secondary Benefits

- Safe collaboration with untrusted parties
pseudonymisation
- Data can be used for secondary purposes without explicit consent of the data subjects
- Reduced regulatory pressure
- Competitive advantage “Our product offers robust privacy measures than others”
- Mitigate the difficulties in obtaining datasets for product testing or model training purposes

PETs

How do PETs achieve data protection

- Transform data or derived data
 - Pseudonymisation
 - K-Anonymity
 - Differential Privacy
 - Synthetic Data
- Hide or shield data
 - Pseudonymisation
 - Homomorphic Encryption
 - Zero-Knowledge Proof
- Split datasets or control access to datasets
 - Pseudonymisation
 - Federated Learning

What about the utility?

PETs

Maturity

- Cutting edge does not mean the technology is applicable
- Standards in application might not have been developed

Ex: <https://homomorphicencryption.org/standard/>

- Some PETs can work better in theoretical settings

- How can we decide the maturity

- ✓ Standard measurement systems

Ex: [Technology Readiness Levels \(TRL\)](#)

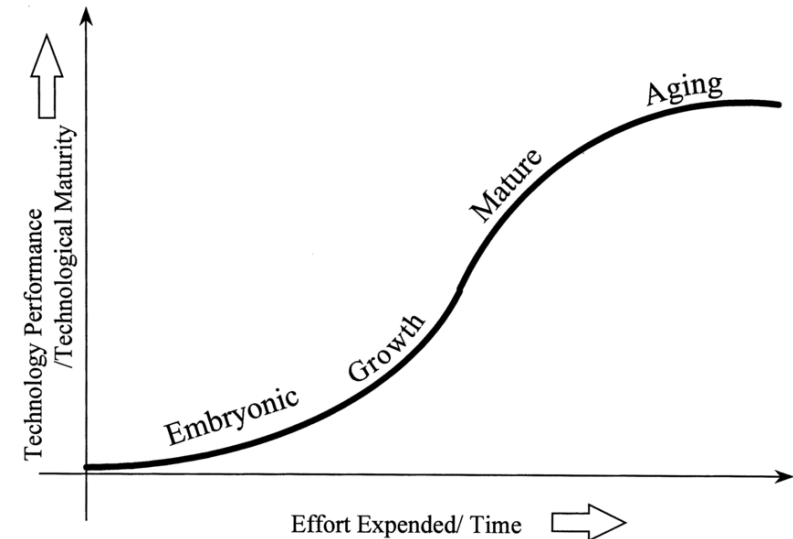
- ✓ Quality measures

Scalability

Versatility

Level of protection

Robustness



Performance analysis of technology using the S curve model: The case of digital signal processing (DSP) technologies (Nieto, Mariano & López, Francisco & Cruz-Roldan, Fernando. (1998).)

PETs

Known Weaknesses

- Pseudonymisation
 - Risk of re-identification

- K-Anonymity
 - Risk of re-identification

- Differential Privacy
 - No consensus over the optimal privacy vs utility tradeoff
 - Performance overhead
 - Require higher computational power
 - Limited chances of gaining insights

- Federated Learning
 - Depends on architectural specifications
 - Data quality and format at the local nodes matter
 - Performance overhead

PETs

Known Weaknesses

- Homomorphic Encryption
 - Performance overhead
 - Limited functionality

- Zero-Knowledge Proof
 - Interactive protocols are vulnerable to side channel attacks
 - Performance overhead

- Synthetic Data
 - May not present outliers in the original dataset
 - Risk of biased data
 - Difficulty to mimic complex data

PETs

Canvas – Which PET(s) Do You Need?

10:00

Start	Stop	Reset	mins: 10	secs: 0	type: None	▼
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Breaktime for PowerPoint by Flow Simulation Ltd.

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PETs

How to choose?

- Involve personal data handling
- Nature of the data processing : *individual level or aggregate analysis?*
- Scope of the data processing : *internal, collaboration, sharing?*
- Size of the data : *“is it worth to apply a selected PET on 5 records?...”*
- Possible threats to data : *“Is the curator trusted?..”*
- Sufficient resources : *time, manpower, computational power, money,.....*
- Maturity

Summary

- Anonymity
- Privacy Enhancing Technologies (PETs)
 - Several PETs
 - Maturity
 - Weaknesses
 - How to choose a PET



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